General Bite #12: Specialized Antennas (Inverted V, EFHW, Loops, Beverages, Screwdrivers, Log-Periodics)

NVIS dipoles, end-fed half-waves, halo, traps, log-periodics, screwdriver mobile antennas, Beverages, electrically small loops.

Group G9D · 12 pool questions · 5-10 minute presentation · Practice exams at eham.net/exams

NVIS Configuration

Most effective antenna for daytime 40 m NVIS: A horizontal dipole placed between 1/10 and 1/4 wavelength above the ground (G9D01).
Low-and-flat is what fires the signal up into the F2 layer for short-range (0-300 mi) HF coverage.

End-Fed Half-Wave

Feed point impedance of an end-fed half-wave antenna: Very high (G9D02).
Voltage maximum at the end → impedance maximum (typically several thousand ohms). Needs a 49:1 unun to match 50 Ω coax (the standard EFHW transformation ratio).

Halo And Stacking

Question	Answer	Q
Maximum radiation from a VHF/UHF halo antenna	**Omnidirectional in the plane of the halo** (G9D03). A halo is a bent half-wave dipole formed into a ring; the circular geometry gives equal radiation in all azimuthal directions.	G9D03
Advantage of vertically stacking horizontally polarized Yagis	**It narrows the main lobe in elevation** (G9D05). Two stacked antennas in the vertical plane sharpen the elevation pattern.	G9D05

Multiband Tricks

Question	Answer	Q
Primary function of antenna traps	**To enable multiband operation** (G9D04). A trap resonates at one band, isolating part of the antenna for higher-band operation.	G9D04
Disadvantage of multiband antennas	**They have poor harmonic rejection** (G9D11). Resonance on multiple bands = harmonics from one band live on a neighboring band.	G9D11
Log-periodic antenna advantage	**Wide bandwidth** (G9D06).	G9D06
What describes a log-periodic antenna?	**Element length and spacing vary logarithmically along the boom** (G9D07).	G9D07

Mobile Whip Tuning

How does a "screwdriver" mobile antenna adjust its feed point impedance? By varying the base loading inductance (G9D08).
A motor extends or retracts a sliding contact along a loading coil, changing the inductance and resonating the short whip on different bands.

Receive-Only Specialty Antennas

Question	Answer	Q
Primary use of a Beverage antenna	**Directional receiving for MF and low HF bands** (G9D09). Long horizontal wire, terminated at the far end.	G9D09
Null direction of an electrically small loop (under 1/10 λ circumference)	**Broadside to the loop** (G9D10). The pattern is a figure-8 with lobes in the plane of the loop; nulls are broadside to the loop, off the flat faces (perpendicular to the plane of the loop).	G9D10

Common Dipole Configuration

Common name of a dipole with a single central support: Inverted V (G9D12). Apex at the center, ends drooping down at 30-60° below horizontal. Convenient because you need only one tall support.

Mental Cheat Sheet

- NVIS 40 m antenna = **horizontal dipole, 1/10 to 1/4 λ high**.
- End-fed half-wave Z = **very high**.
- Halo radiation = **omni in the plane of the loop**.
- Stacked horizontal Yagis → narrows **elevation** lobe.
- Antenna traps → **multiband** operation.

Common Traps

- NVIS antenna is **low** (1/10-1/4 λ), not at 1/2 λ height.
- End-fed half-wave Z is **very high**, not 50 Ω.
- Halo is omni **in the plane of the halo**, not "broadside."
- Stacked Yagis narrow **elevation**, not "horizontal beamwidth" or "azimuth."

- Multiband antenna disadvantage = **poor harmonic rejection**.
- Log-periodic advantage = **wide bandwidth**.
- Log-periodic: lengths/spacing vary **logarithmically**.
- Screwdriver antenna = motor-varied **base loading inductance**.
- Beverage = directional receive on **low HF / MF**.
- Small loop nulls = **broadside to the loop**.
- Single-mast dipole = **inverted V**.

- Traps enable **multiband**, not "notch spurious frequencies."
- Multiband downside is **harmonic rejection**, not "must use antenna tuner."
- Log-periodic = **wide BW**, not "higher gain per element than Yagi."
- Screwdriver tunes **inductance**, not "body capacitance" or "extends the whip."
- Beverages are **receive-only** for low HF, not for higher HF or transmit.
- Small loop nulls are **broadside** (perpendicular), not "in the plane of the loop."
- Single-support dipole = **Inverted V**, not "Inverted L" (that's a different antenna).

Takeaway: NVIS antennas are low and horizontal. End-fed half-waves see very high impedance. Halos go omni in their plane. Traps enable multibanding at the cost of harmonic rejection. Log-periodics get wide bandwidth from log-spaced elements. Screwdriver mobile antennas tune by varying base inductance. Beverages are long, directional, receive-only antennas for low HF. Small loops null broadside.

Bite 12 Question Bank (G9D)

G9D01. Which of the following antenna types will be most effective as a near vertical incidence skywave (NVIS) antenna for short-skip communications on 40 meters during the day?
A. A horizontal dipole placed between 1/10 and 1/4 wavelength above the ground.
B. A vertical antenna placed between 1/4 and 1/2 wavelength above the ground.
C. A horizontal dipole placed at approximately 1/2 wavelength above the ground.
D. A vertical dipole placed at approximately 1/2 wavelength above the ground.

G9D02. What is the feed point impedance of an end-fed half-wave antenna?
A. Very low.
B. Approximately 50 ohms.
C. Approximately 300 ohms.
D. Very high.

G9D03. In which direction is the maximum radiation from a VHF/UHF "halo" antenna?
A. Broadside to the plane of the halo.
B. Opposite the feed point.
C. Omnidirectional in the plane of the halo.
D. On the same side as the feed point.

G9D04. What is the primary function of antenna traps?
A. To enable multiband operation.
B. To notch spurious frequencies.
C. To provide balanced feed point impedance.
D. To prevent out-of-band operation.

G9D05. What is an advantage of vertically stacking horizontally polarized Yagi antennas?
A. It allows quick selection of vertical or horizontal polarization.
B. It allows simultaneous vertical and horizontal polarization.
C. It narrows the main lobe in azimuth.
D. It narrows the main lobe in elevation.

G9D06. Which of the following is an advantage of a log-periodic antenna?
A. Wide bandwidth.
B. Higher gain per element than a Yagi antenna.
C. Harmonic suppression.
D. Polarization diversity.

G9D07. Which of the following describes a log-periodic antenna?
A. Element length and spacing vary logarithmically along the boom.
B. Impedance varies periodically as a function of frequency.
C. Gain varies logarithmically as a function of frequency.
D. SWR varies periodically as a function of boom length.

G9D08. How does a "screwdriver" mobile antenna adjust its feed point impedance?
A. By varying its body capacitance.
B. By varying the base loading inductance.
C. By extending and retracting the whip.
D. By deploying a capacitance hat.

G9D09. What is the primary use of a Beverage antenna?
A. Directional receiving for MF and low HF bands.
B. Directional transmitting for low HF bands.
C. Portable direction finding at higher HF frequencies.
D. Portable direction finding at lower HF frequencies.

G9D10. In which direction or directions does an electrically small loop (less than 1/10 wavelength in circumference) have nulls in its radiation pattern?
A. In the plane of the loop.
B. Broadside to the loop.
C. Broadside and in the plane of the loop.
D. Electrically small loops are omnidirectional.

G9D11. Which of the following is a disadvantage of multiband antennas?
A. They present low impedance on all design frequencies.
B. They must be used with an antenna tuner.
C. They must be fed with open wire line.
D. They have poor harmonic rejection.

G9D12. What is the common name of a dipole with a single central support?
A. Inverted V.
B. Inverted L.
C. Sloper.
D. Lazy H.

Answer Key

G9D01: A	G9D02: D	G9D03: C	G9D04: A
G9D05: D	G9D06: A	G9D07: A	G9D08: B
G9D09: A	G9D10: B	G9D11: D	G9D12: A